

- 7 Let $\{f_n\}$ be a sequence in $L^1(X, \mathcal{M}, \mu)$ such that $f_n \rightarrow f$ uniformly, $\mu(X) < \infty$. We want to show that $f \in L^1$ and $\lim \|f - f_n\|_1 = 0$. Note that for some $\epsilon > 0$ there exists N_ϵ such that for all $n \geq N_\epsilon$ we have that $|f - f_n| < \frac{\epsilon}{\mu(X)}$. Therefore $\int_X |f - f_n| d\mu < \frac{\epsilon}{\mu(X)} \int_X d\mu = \epsilon$. Therefore in the limit it goes to zero. Additionally $\int_X |f| d\mu \leq \int_X |f - f_n| d\mu + \int_X |f_n| d\mu < \infty + \epsilon = \infty$, giving us that $f \in L^1$. Note that this fails to hold if $\mu(X) = \infty$. In particular if one takes the functions $f_n(x) = \sum_{k=1}^{\infty} \frac{1}{k^{1+\frac{1}{n}}} 1_{(k-1, k)}(x)$, each $\int_{\mathbb{R}} |f_n| d\mu$ is guaranteed to converge by the p-series tests. However if one takes n to infinity you get the harmonic series, which diverges.
- 8 Let $f(x) = x^{-1/2} 1_{(0,1)}$, $\{q_n\}_{n \in \mathbb{N}}$ be an enumeration of the rationals, and $g(x) = \sum_{n=1}^{\infty} f(x - q_n)$. We claim that $g \in L^1$. Note that first each $2^{-n} f(x - q_n)$ is an integrable function, therefore we can interchange the sum and the integral: $\int_{\mathbb{R}} |g| d\mu = \int_{\mathbb{R}} \sum_{n=1}^{\infty} 2^{-n} f(x - q_n) d\mu = \sum_{n=1}^{\infty} \int_{\mathbb{R}} 2^{-n} f(x - q_n) d\mu$. Furthermore, by the construction of f , $\int_{\mathbb{R}} f(x - q_n) d\mu = \int_{\mathbb{R}} f d\mu$, since the shifting of both the function and the indicator exactly cancel out. Therefore $\int_{\mathbb{R}} |g| = \sum_{n=1}^{\infty} 2^{-n} \int_{\mathbb{R}} f d\mu = \sum_{n=1}^{\infty} 2^{1-n} = 1$. We additionally claim for all intervals $I \subset \mathbb{R}$, $\lambda \geq 1$ that $\mu(I \cap \{x : g(x) > \lambda\}) > 0$. Let $(a, b) = I$, $\lambda \geq 1$, and let q_a, q_b be rational numbers such that $a < q_a < q_b < b$ and $q_b - q_a < 1$. If we then consider $2^{-\alpha} f(x - \frac{q_b + q_a}{2})$ where α is the index of $\frac{q_b + q_a}{2}$ we are guaranteed that g has a singularity contained within said interval. Furthermore, the interval $(\frac{q_b + q_a}{2}, \min(\frac{1}{\frac{q_b + q_a}{2} + \lambda^2}, q_b))$ is guaranteed to be of non-zero measure. Thus $\mu(I \cap \{x : g(x) > \lambda\}) > 0$. Note that since g is finite almost everywhere, then on that same set g^2 is finite. However by cauchy-schwarz we have that

$$\sum_{n=1}^{\infty} 4^{-n} f(x - q_n)^2 \leq g^2.$$

Since for a single n , we have that $\int_{\mathbb{R}} f(x - q_n)^2 = \int_0^1 \frac{1}{x} dx = \infty$, and since the rationals are dense in $\mathbb{R}$ then every interval will necessarily contain a term which diverges, thus making it not integrable on said interval.

- 9 Let $f_n(x) = 1_{E_n}(x)$ be a cauchy sequence of indicators in $L^1(X, \mathcal{M}, \mu)$. Note by the completion of L^1 then there exists $f \in L^1$ such that $\lim f_n = f$ almost everywhere. Note that since the set E where they disagree is a set of measure zero, we can simply multiple f by 1_{E^c} , which ensures that $f 1_{E^c}$ is an indicator function. As on E^c , f is an indicator function since it can only be 1 or 0. On E if we set all values to 0 then it still is an indicator function. Since they differ only on a set of measure 0 then the sequence converges to an indicator function in L^1 .

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- 11 Let t_n be some sequence converging to $t > 0$. Note that $f(x)g(t_n x) \leq M|f(x)|$ for all $x \in \mathbb{R}$ and $\int_{\mathbb{R}} M|f(x)| d\mu < \infty$. Thus by LDCT we have that $\lim \int_{\mathbb{R}} f(x)g(t_n x) d\mu = \int_{\mathbb{R}} \lim f(x)g(t_n x) d\mu$. Furthermore any constant positive "multiple" of the set is still of measure zero. Therefore $\lim g(t_n x) = g(tx)$. Thus $\lim \int_{\mathbb{R}} f(x)g(t_n x) d\mu = \int_{\mathbb{R}} f(x)g(tx) d\mu = \int_{\mathbb{R}} f(x)g(t x) d\mu$. Thus F is continuous.

- 12 Let $(X, \mathcal{M}, \mu)$ be a finite measure space, $\{f_n\}$ be a sequence of integrable functions such that $f_n \rightarrow 0$ in measure and $\sup \| |f_n|^2 \|_1 < \infty$. We claim that the sequence converges to 0 in L^1 . Let $\epsilon > 0$ and $E_{n,\epsilon} = \{x : |f(x)| \geq \epsilon\}$. Note that $\lim \mu(E_{n,\epsilon}) = 0$ for all ϵ and that

$$\begin{aligned} \left| \int_X f_n d\mu \right|^2 &\leq \left| \int_{E_{n,\epsilon}} f_n d\mu \right|^2 + \left| \int_{E_{n,\epsilon}^c} f_n d\mu \right|^2 + \left| \int_{E_{n,\epsilon}} f_n d\mu \right| \left| \int_{E_{n,\epsilon}^c} f_n d\mu \right| \\ &\leq \left| \int_{E_{n,\epsilon}} f_n d\mu \right|^2 + \epsilon \mu(E_{n,\epsilon}) \left| \int_{E_{n,\epsilon}^c} f_n d\mu \right| + \epsilon^2 \mu(E_{n,\epsilon}^c) \\ &\leq \mu(E_{n,\epsilon}) \int_X |f_n|^2 d\mu + \epsilon \mu(E_{n,\epsilon}) \left| \int_{E_{n,\epsilon}^c} f_n d\mu \right| + \epsilon^2 \mu(E_{n,\epsilon}^c) \end{aligned}$$

Since $\sup \| |f_n|^2 \|_1 < \infty$, then let $M = \sup \| |f_n|^2 \|_1$. Additionally, since $\lim \mu(E_{n,\epsilon}) = 0$ then $\lim \int_{E_{n,\epsilon}} f_n d\mu = 0$ as an integral on a set of measure zero is zero. Therefore the inequality is bounded by $\epsilon^2 \mu(E_{n,\epsilon}^c)$, therefore the integral goes to zero.

- 13 Let $(X, \mathcal{M}, \mu)$ be a measure space, μ is a semifinite measure, f is a non-negative function measurable function such that for all $g \in L^+(X, \mathcal{M}, \mu)$,

$$\int_X fg d\mu < \infty.$$

We claim that f is uniformly bounded almost everywhere. Suppose not. Then for all $M > 0$ there exists a set E_M such that for all $x \in E_M$, $f(x) > M$